

Assignment 2: Visualization by example

Zeynep Cakez

2024-09-07

Exercise 1

- i. How many rows and columns does this dataset have?

The dataset has 344 rows and 8 columns.

- ii. What does a row in this dataset represent (i.e. what is the unit of observation)?

It represents one penguin.

- iii. What are three categorical variables in the penguins dataset?

Species, island, and sex.

- iv. What are four continuous variables in the penguins dataset?

Bill length, bill depth, flipper length, and mass.

- v. Which variable in the penguins dataset could be treated as either continuous or categorical, depending on the context in which it is used?

Year, because it is numeric and it isn't too varied.

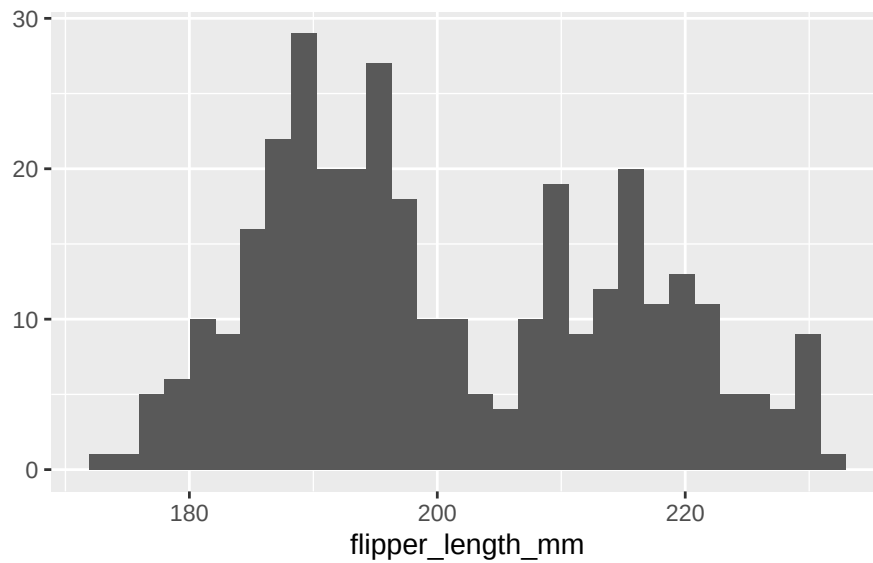
- vi. What are the three species of penguin in the dataset?

Adelie, Gentoo, and Chinstrap.

Exercise 2

```
qplot(x = flipper_length_mm, data = penguins)
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



i. Which axis of the plot has the flipper length variable been plotted on?

The x-axis.

ii. What do the numbers on the other axis of the graph represent?

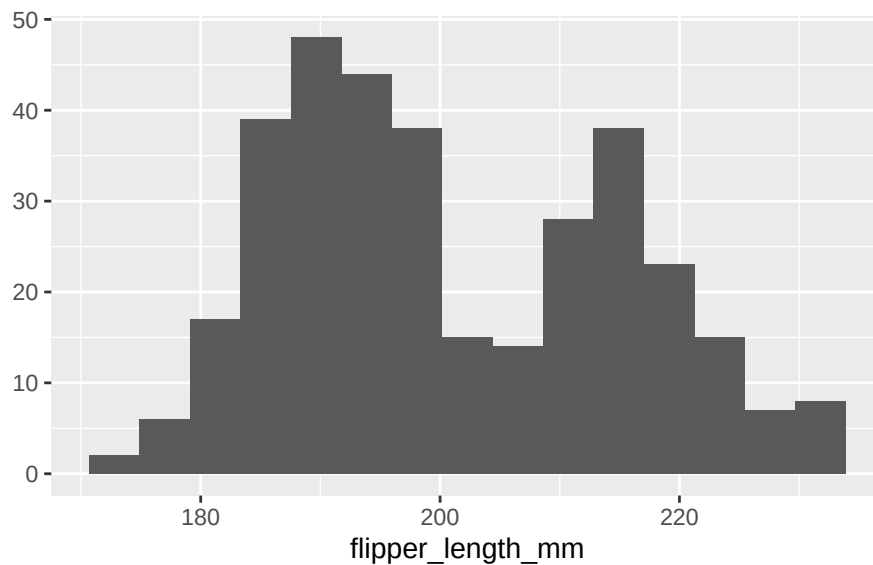
The amount of penguins.

iii. What is the modality of the distribution of flipper lengths?

Bimodal.

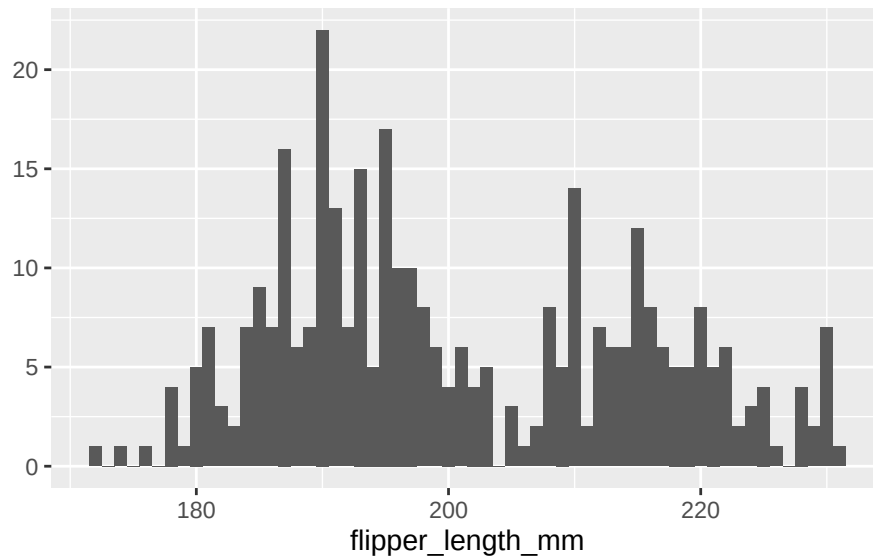
Exercise 3

```
qplot(x = flipper_length_mm, bins = 15, data = penguins)
```



Cleaner than Exercise 1 graph.

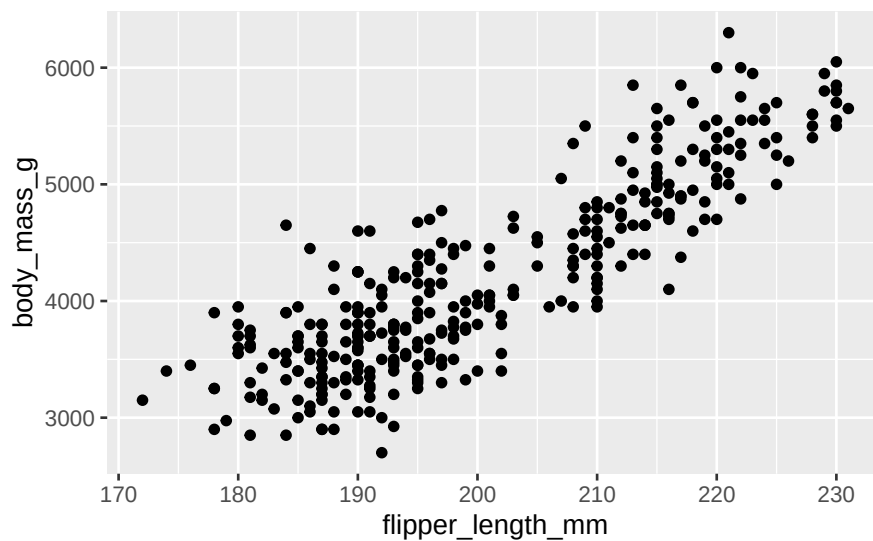
```
qplot(x = flipper_length_mm, binwidth = 1, data = penguins)
```



Messier than Exercise 2 graph.

Exercise 4

```
qplot(x = flipper_length_mm, y = body_mass_g, data = penguins)
```



- i. What variable is on the y-axis of this scatter plot?

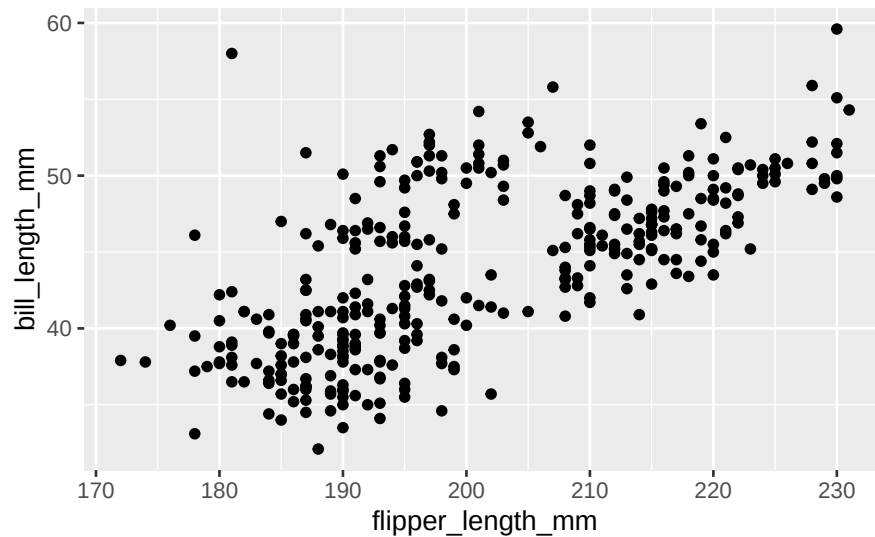
The mass of the penguins.

- ii. Is there a relationship between these two variables, and if so, is it linear or non-linear?

Yes, there is. The graph is positive and linear.

Exercise 5

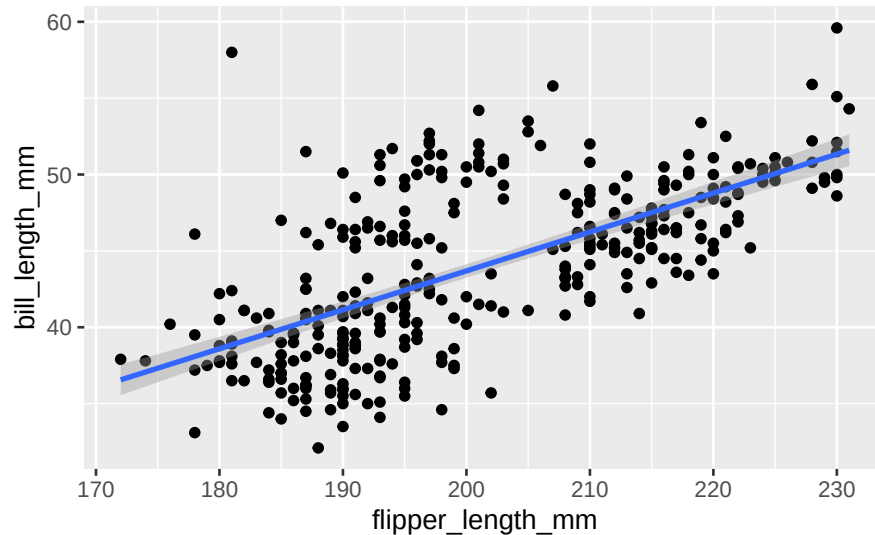
```
qplot(x = flipper_length_mm, y = bill_length_mm, data = penguins)
```



This graph shows a linear relationship; however, there is a low correlation between flipper length and bill length.

Exercise 6

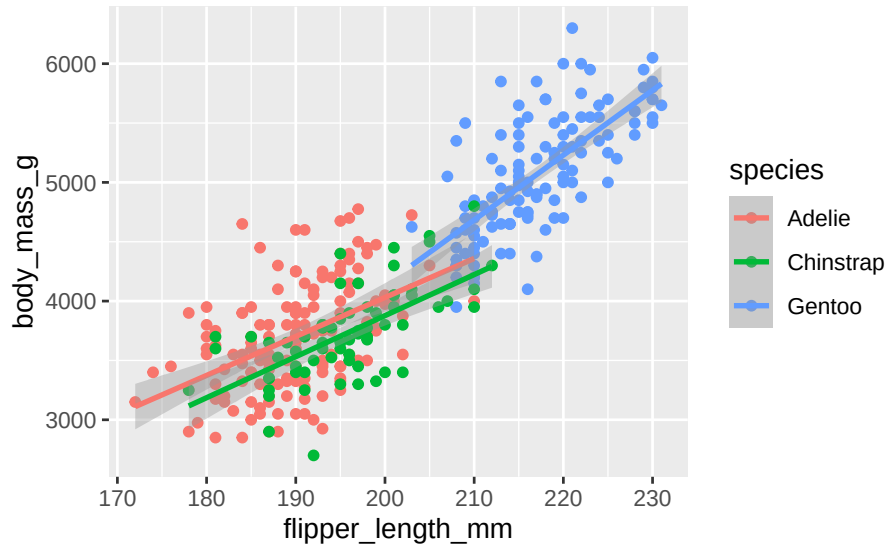
```
qplot(x = flipper_length_mm, y = bill_length_mm, data = penguins, method = "lm", geom = c("point", "smooth"))  
## `geom_smooth()` using formula = 'y ~ x'
```



If you take out the scatter points and just show the trend line, it shows a simpler version of the relationship between body mass and bill length but doing that can hide important data.

Exercise 7

```
qplot(x = flipper_length_mm, y = body_mass_g, data = penguins, color = species, method = "lm",  
      ## `geom_smooth()` using formula = 'y ~ x'
```



Coloring the data points by species makes it easy to see how each group of penguins has its own pattern in the relationship between bill length and body mass.